

IVDEL', Russian Federation

WMO: 239210

Lat: 60.683N

Lon: 60.450E

Elev: 94

StdP: 100.20

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.5	-33.8	-39.3	0.1	-36.4	-36.6	0.1	-33.7	6.2	-7.7	5.3	-9.0	0.7	320	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	29.0	19.9	27.0	18.9	24.9	17.5	21.0	26.8	19.8	25.6	18.6	23.6	2.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	23.6	17.7	12.8	22.6	16.6	11.9	21.2	61.2	26.9	57.2	25.7	53.2	23.6	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.2	4.4	DB	-39.5	31.6	3.2	1.6	-41.8	32.7	-43.7	33.7	-45.5	34.6	-47.8	35.8	(4)
(5)			WB	-39.5	22.8	3.2	1.4	-41.8	23.8	-43.7	24.6	-45.5	25.4	-47.8	26.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-18.6	-15.7	-6.2	1.6	8.4	14.5	17.3	13.9	8.1	1.3	-9.3	-15.6	(6)
(7)		DBStd	13.76	8.27	8.15	6.05	5.08	4.80	4.51	3.96	3.93	4.21	4.81	8.67	9.26	(7)
(8)		HDD10.0	4206	887	719	501	254	92	12	2	8	86	271	581	795	(8)
(9)		HDD18.3	6734	1145	953	759	502	310	130	69	147	308	527	831	1053	(9)
(10)		CDD10.0	574	0	0	0	2	41	147	226	128	28	2	0	0	(10)
(11)		CDD18.3	62	0	0	0	0	1	16	36	9	1	0	0	0	(11)
(12)		CDH23.3	778	0	0	0	0	42	201	413	112	11	0	0	0	(12)
(13)		CDH26.7	192	0	0	0	0	5	45	111	29	1	0	0	0	(13)
(14)	Wind	WSAvg	1.7	1.3	1.5	2.0	2.2	2.1	1.8	1.5	1.6	1.9	1.6	1.4	(14)	
(15)	Precipitation	PrecAvg	547	27	19	28	34	54	71	72	76	71	39	32	23	(15)
(16)		PrecMax	723	50	39	71	56	106	162	153	168	160	74	57	43	(16)
(17)		PrecMin	347	5	3	1	0	11	21	21	32	7	10	2	6	(17)
(18)		PrecStd	79	12	10	19	14	26	40	31	40	40	20	15	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	3.5	9.1	19.8	27.3	30.1	31.5	30.2	24.8	16.2	6.6	3.8	(19)
(20)			MCWB	-0.5	0.6	3.5	11.3	15.9	19.4	21.0	21.3	17.0	10.8	4.0	1.2	(20)
(21)		2%	DB	-1.9	0.9	6.1	15.5	24.1	27.9	29.6	26.2	20.0	12.9	4.5	1.5	(21)
(22)			MCWB	-3.3	-1.1	2.0	8.4	13.8	18.3	20.4	18.8	14.1	9.4	2.4	-0.4	(22)
(23)		5%	DB	-4.0	-1.3	4.2	12.6	21.3	25.6	27.7	23.5	17.1	10.3	2.9	-0.6	(23)
(24)			MCWB	-5.2	-3.1	0.6	6.5	12.6	17.5	19.7	17.6	12.9	7.5	0.9	-2.1	(24)
(25)		10%	DB	-6.3	-3.5	2.5	10.0	18.4	23.2	25.7	21.1	15.0	8.0	1.2	-2.9	(25)
(26)			MCWB	-7.4	-5.0	-0.4	5.0	11.1	16.1	18.9	16.4	11.6	5.9	-0.4	-4.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.4	0.8	4.2	11.6	16.5	21.3	22.9	21.9	17.6	11.6	4.4	1.3	(27)
(28)			MCDWB	0.9	3.0	8.1	18.9	24.8	26.9	28.2	27.4	23.3	14.8	6.1	3.2	(28)
(29)		2%	WB	-3.2	-0.9	2.3	8.9	14.8	19.6	21.5	19.7	14.9	9.4	2.7	-0.3	(29)
(30)			MCDWB	-2.2	0.9	5.3	14.6	22.2	25.7	27.3	25.2	18.8	12.3	4.3	1.5	(30)
(31)		5%	WB	-5.1	-2.9	1.0	7.0	13.3	18.4	20.6	18.2	13.3	7.7	0.9	-2.0	(31)
(32)			MCDWB	-4.0	-1.3	3.8	12.1	19.9	24.1	26.3	22.3	16.6	10.1	2.5	-0.6	(32)
(33)		10%	WB	-7.4	-5.0	-0.2	5.3	11.8	17.1	19.6	16.9	12.0	6.1	-0.4	-4.1	(33)
(34)			MCDWB	-6.3	-3.5	2.4	9.2	17.5	21.9	24.9	20.5	14.7	7.9	1.1	-3.0	(34)
(35)	Mean Daily Temperature Range	MDBR		9.0	10.8	11.0	10.2	11.9	10.9	11.0	9.8	8.6	6.5	6.9	8.2	(35)
(36)		5% DB	MCDBR	8.9	10.0	11.5	15.6	17.9	14.9	14.8	14.1	12.8	10.3	6.3	7.4	(36)
(37)			MCWBR	8.0	8.3	8.1	9.3	9.0	6.9	7.0	7.4	7.4	6.9	5.1	6.4	(37)
(38)		5% WB	MCDWB	8.8	9.4	10.3	14.5	15.9	13.1	13.3	13.0	11.4	9.6	6.0	7.4	(38)
(39)			MCWBR	8.0	8.0	7.6	9.1	8.6	6.7	6.9	7.4	6.8	6.8	5.0	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.254	0.294	0.343	0.379	0.380	0.385	0.413	0.407	0.353	0.319	0.268	0.237	(40)
(41)		taud		2.021	2.069	2.033	2.133	2.340	2.384	2.304	2.297	2.422	2.363	2.119	1.954	(41)
(42)		Ebn at Noon		465	667	753	805	843	844	805	776	764	659	450	311	(42)
(43)		Edh at Noon		44	78	114	125	111	108	114	107	79	60	40	29	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.36	1.15	2.34	3.67	4.89	5.14	5.22	3.78	2.17	1.08	0.41	0.19	(44)
(45)		RadStd		0.03	0.12	0.20	0.46	0.43	0.42	0.43	0.36	0.32	0.13	0.05	0.02	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page